plaintext, you use the 2nd character of the keyword and so on. Hence, the letters of the keyword are used in a cyclic manner. Hence if you were to encrypt the complete plaintext using the given keyword, it would become: BCOXOWZGKDFRGCJBEE. So? Is the ciphertext now traceable back to the original plaintext? Well, no. Not at least without the keyword. This brings us to two interesting conclusions:

The strength of the keyword would determine how the encrypted text turns out. For example, if we chose a small keyword such as 'HIM' or 'ME', it would be much more easier to track it back to the original plaintext. Not only the length of the keyword but also the randomness of the characters in the keyword improve the strength of the cipher. For example, if we used a keyword such as 'DDDDDD', would it increase the strength of the cipher? No, it would actually weaken the cipher as it would just boil down to 'D' being used as a keyword which would directly translate to a rotation of 3 on the alphabet – pretty weak, isn't it? On the other hand, if we use "CRAZYFOX" as the keyword, it strengthens the cipher because it's longer and has no repeating characters. To use "XARCZYFO" would be even better because this makes the keyword itself gibberish and more difficult to predict. These principles (longer keyword, random character distribution in keyword and non-dictionary keywords) still happen to be important factors governing the strength of the encryption.

Spaces in the plaintext could make the ciphertext weaker. If you notice, our plaintext didn't have any white-spaces between them. If they had, the ciphertext would contain spaces too, or some other character to represent white spaces which make the text more guessable than when it does not.

Transposition Cipher

The word says it all – it uses transposition techniques to encrypt the plaintext. The word 'transposition' means to change the position. This cipher, however doesn't work on character level, but on the word level. In this case we do not change the letters themselves but the way we read them. Let's take our previous example with "I LOVE DIGIT MAGAZINE" as our plaintext (alright, we added spaces). When we write the same sentence backwards, it becomes "ENIZAGAM TIGID EVOL I". This is certainly more difficult to figure out than the plaintext but breaking it's not a big deal by any means. If we were to eliminate the spaces, it would be "ENIZAGAMTIGIDEVOLI", making it a bit more difficult. There are other more popular transposition ciphers which include: